

RESUME

Daeki Hong

Senior Unity/XR Systems Engineer

I have spent ten years building industrial XR, BIM runtimes, digital-twin authoring tools, robotics interfaces, and VR simulators in Unity C#. Because I also install and hand over the systems, I check large-model performance, real-time input, hardware synchronization, and failure recovery early.

Unity C# Runtime

Industrial XR

BIM Geometry

Robotics / IK

Editor Tooling

Field Delivery

CONTACT

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Location Seoul, Korea

BEST FIT

Industrial Unity simulation, XR training, BIM/digital-twin runtime, robotics control interfaces, EditorWindow tooling, and field-delivered software.

EDUCATION

B.S. in Software Engineering, Korea Aerospace University.

SELECTED WORK

UNITY INDUSTRIAL DELIVERY

VR Robot / DXCenter

Built a Quest 3 - Unity - Flask - RoboDK - Doosan A0912 remote operation flow and a Unity 6 URP/WebGL digital-twin authoring tool, including safety logic, calibration, validation, and non-developer EditorWindow tooling.

PROBLEMS SOLVED DIRECTLY

IK / BIM / Hardware Sync

I implemented a Jacobian IK solver, runtime BIM section cuts, nine EditorWindow tools, and hardware-to-Unity synchronization when existing tools did not meet the delivery requirement.

BIM RUNTIME

WatchBIM / Unity-CrossSection

Delivered custom hit detection, mesh optimization, auto LOD, runtime cross-sections, 2D analysis views, slope labels, minimap, and camera tooling for large road BIM data.

FIELD DELIVERY

VR Simulator Collection

Built construction, factory, naval, FPS safety, and AR training applications with hardware integration, multiplayer mission flow, installation, operation, removal, and operator handoff.

EXPERIENCE

Unity Engineering and Field Delivery

YMX - Senior Engineer, Client Team 2

2024.12 - PRESENT

Industrial Unity development: digital-twin authoring tools, robotics interfaces, and exhibition systems.

- Built a Unity 6 URP/WebGL DXCenter authoring environment with nine EditorWindow tools, ScriptableObject data flow, Service Locator, EventBus, strategy-based state effects, adapter-style device integration, validation flow, and runtime preparation for non-developers.
- I passed Quest 3 input through Unity and a Flask bridge to a Doosan A0912. I implemented coordinate conversion, smoothing, speed and acceleration limits, TCP/HTTP communication, calibration, and safety handling, then supported the public BEXCO demo.
- Implemented an internal Jacobian IK solver to reduce RoboDK dependency, including 6-DOF tracking, damping near singularities, debug UI, and multi-robot validation.

Jangheon Industrial - Seoul Office

2020.06 - 2024.12

BIM visualization, geometry tooling, drawing tools, and engineering software.

- Developed WatchBIM runtime features for highway BIM data, including custom hit engine, mesh optimization, auto LOD, runtime cross-section, 2D analysis viewer, slope label, minimap, and camera tools.
- Built Unity-CrossSection as a reusable C# module using coroutines, NativeArray, Burst-compatible jobs, progress/cancel flow, mesh-plane intersection, caching, and shader visualization.
- Developed road-design drawing tools with WPF/MVVM, MySQL, and CAD entity creation.

Mline Studio / Saltworks / Swink

2015 - 2019

Unity gameplay, VR simulators, Android AR safety training, RS232 device integration, multiplayer mission flow, field installation, operation, removal, and operator handoff.

- Developed construction, factory, and naval safety training VR, FPS safety education, AR education apps, and hardware-linked XR content, then supported customer-site delivery and operator handoff.

UNITY FOCUS

RUNTIME

C# / Unity Runtime

Real-time input, coordinate conversion, state flow, camera control, simulation loops, and operator feedback inside Unity runtime.

TOOLING

EditorWindow Workflow

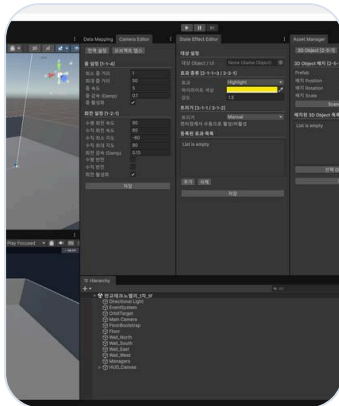
Editor tooling for data mapping, validation, asset registration, and runtime preparation without requiring code changes.

DELIVERY

Hardware / Field

RS232, robot/device links, installation, demos, safety handling, and operator handoff included in the delivery scope.

DXCenter



UNITY 6 URP / WEBGL / EDITORWINDOW TOOLING

A digital-twin authoring tool where content creators and operators handle data mapping, validation, device setup, and runtime preparation without editing code.

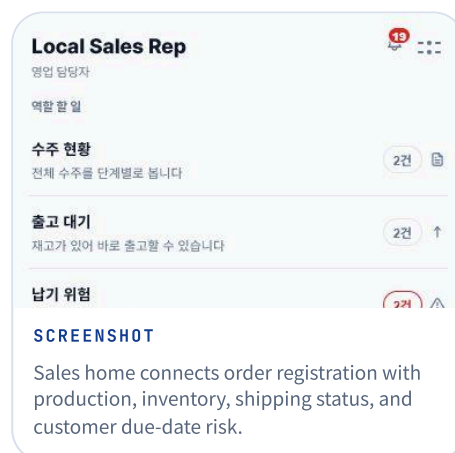
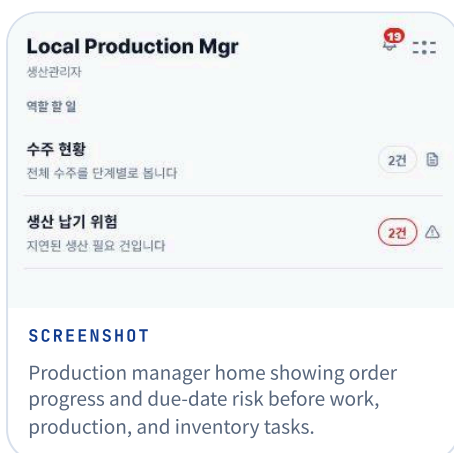
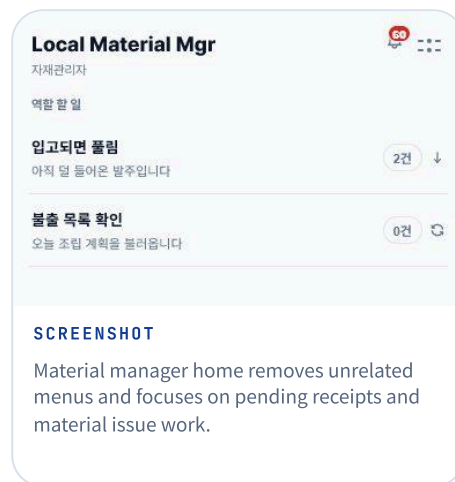
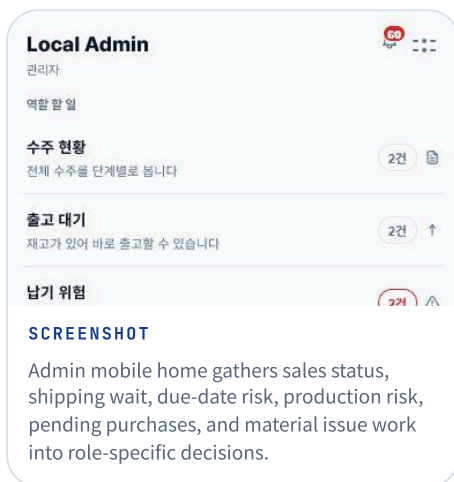
- Designed nine EditorWindow tools around data mapping, validation, asset registration, and runtime readiness.
- Used ScriptableObject data flow, Service Locator, EventBus, strategy-based state effects, and adapter-style device integration.
- I built EditorWindow tools so operators could prepare content without writing code.

SEMS ERP

MOBILE-FIRST ERP / FASTIFY TYPESCRIPT / MYSQL / GCP

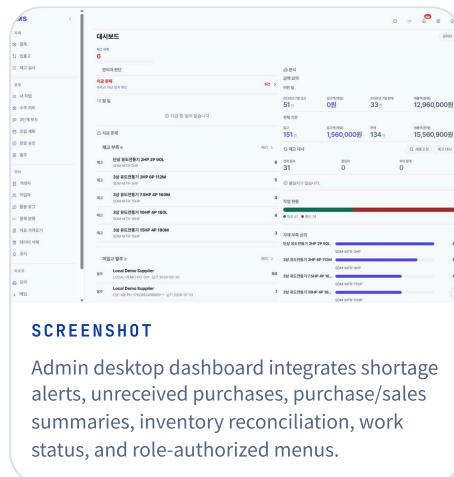
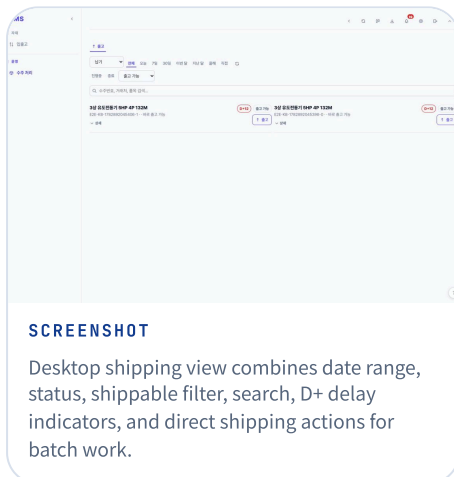
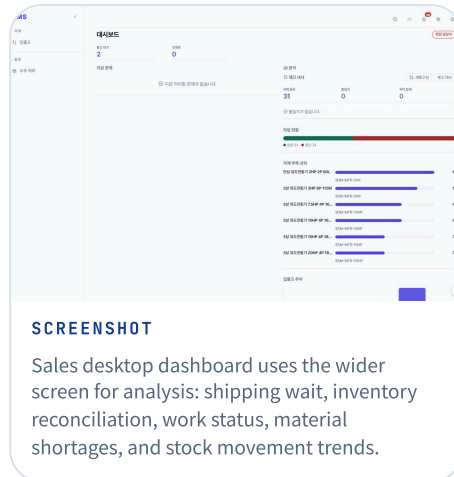
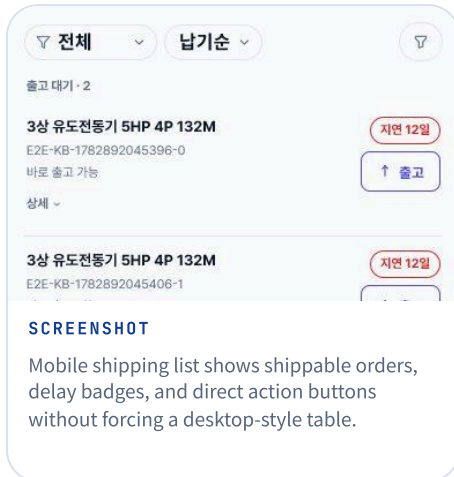
A role-based ERP for small manufacturing teams. Sales orders, BOM, materials, production, outsourcing, inventory, approvals, and shipping are handled in role-specific screens, and shop-floor users can finish today's tasks from a phone.

- Mapped the work of admin, material manager, production manager, part leader, shop-floor user, and sales into separate task screens.
- Built mobile task screens that show each role its next action first.
- Implemented the system around Fastify/TypeScript, MySQL/Kysely, Zod validation, Identity Platform auth, Cloud Run, Cloud SQL, Cloud Storage, tests, and mobile/desktop screenshot checks.



SEMS ERP - additional screenshots 1-4 of 4

A role-based ERP for small manufacturing teams. Sales orders, BOM, materials, production, outsourcing, inventory, approvals, and shipping are handled in role-specific screens, and shop-floor users can finish today's tasks from a phone.



VR Robot Teleoperation

QUEST 3 / UNITY VR / FLASK / ROBODK / DOOSAN A0912

I used a Meta Quest 3 to control a Doosan industrial robot in real time. Unity handled VR input and safety filters, a Python bridge talked to RoboDK, and I later added an in-house IK solver.

- Mapped 6DOF VR controller input into robot target commands with coordinate conversion, smoothing, and safety limits.
- Built the Unity-side feedback loop and bridge assumptions needed for a live BEXCO demonstration.
- Extended the stack with a Jacobian IK solver to reduce external dependency and inspect solver behavior directly.



SCREENSHOT

Expo-side view of the remote operation scenario, including the robot and operator environment.



SCREENSHOT

VR controller input mapped into the IK target so hand motion can drive the robot end-effector.



SCREENSHOT

TCP tracking view showing end-effector target following and solver response.

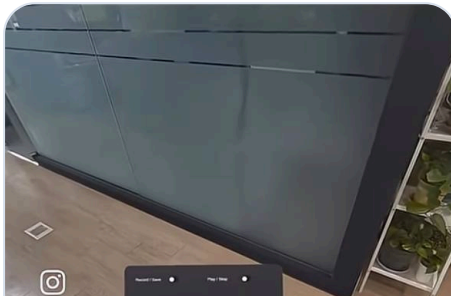


SCREENSHOT

Damping behavior around unstable poses, used to keep the solver usable near singularities.

VR Robot Teleoperation - additional screenshots 1-2 of 2

I used a Meta Quest 3 to control a Doosan industrial robot in real time. Unity handled VR input and safety filters, a Python bridge talked to RoboDK, and I later added an in-house IK solver.



SCREENSHOT

Debug UI for inspecting joints, angles, target error, and solver state during tuning.



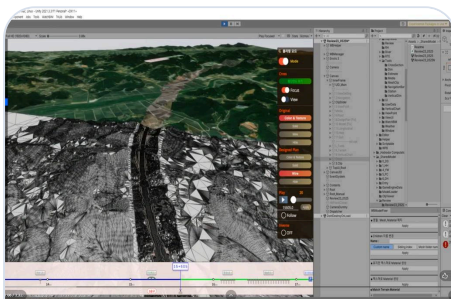
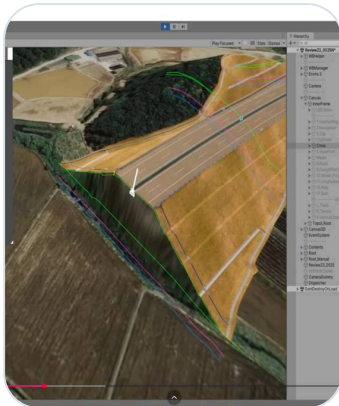
SCREENSHOT

Multi-robot validation scene used to check that the control model was not hard-coded to one arm.

UNITY BIM RUNTIME / GEOMETRY TOOLING / ANALYSIS VIEWER

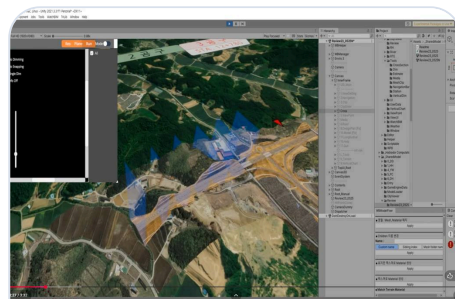
A Unity BIM runtime where operators select large models, cut sections, inspect linked 2D views, and navigate with custom camera tools.

- Built custom hit detection, mesh optimization, auto LOD, and model interaction flows for large BIM data.
- Implemented runtime cross-sections and linked 2D analysis views for operator checks.
- Added camera/minimap/graph tooling so the BIM model could be navigated and inspected in practice.



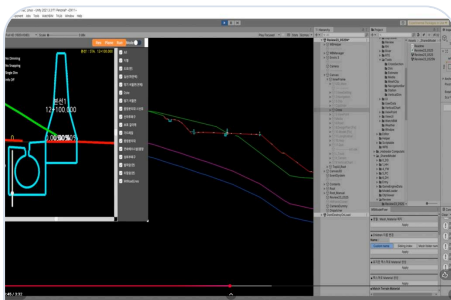
SCREENSHOT

Clipping and selection workflow for navigating large BIM models without relying on external authoring tools.



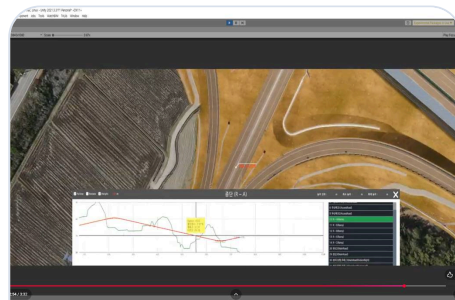
SCREENSHOT

Highlighted section geometry used to make runtime cuts visually clear.



SCREENSHOT

2D analysis viewer linked to the runtime section result for measurement and review.



SCREENSHOT

Vertical graph view for checking elevation and structural distribution.

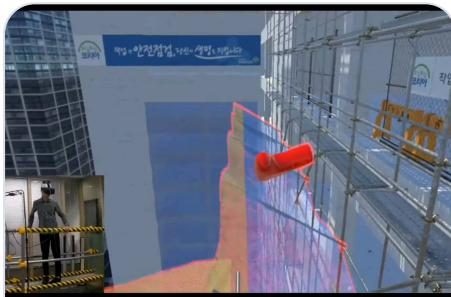
VR Simulator Collection



CONSTRUCTION / FACTORY / NAVAL TRAINING SYSTEMS

Field-delivered VR and AR training projects across construction, factory, and naval scenarios. These projects show installation work, device handling, scripted training flow, and operator handoff.

- Implemented safety training scenes, mission progression, event triggers, and feedback loops in Unity.
- Handled device integration and field constraints such as installation, operation, teardown, and customer handoff.
- Built scenarios where timing, spatial clarity, and training intent mattered more than visual decoration.



SCREENSHOT

Falling accident scenario for construction-site hazard training.



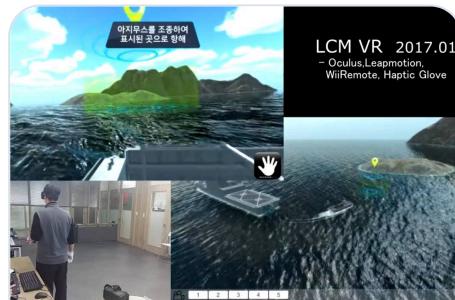
SCREENSHOT

Scaffold scenario focused on spatial awareness and unsafe-work recognition.



SCREENSHOT

Landing craft cockpit simulator view with operator controls and training instrumentation.



SCREENSHOT

Steering interface training for naval simulator operation.

VR Simulator Collection - additional screenshots 1-4 of 4

Field-delivered VR and AR training projects across construction, factory, and naval scenarios. These projects show installation work, device handling, scripted training flow, and operator handoff.



SCREENSHOT

Factory safety training scene for LG Hausys, centered on operator awareness.



SCREENSHOT

Factory accident scenario used to teach hazard response and procedure memory.



SCREENSHOT

Maintenance training context with equipment-state and procedure cues.



SCREENSHOT

Printer/equipment scenario showing the industrial training environment.

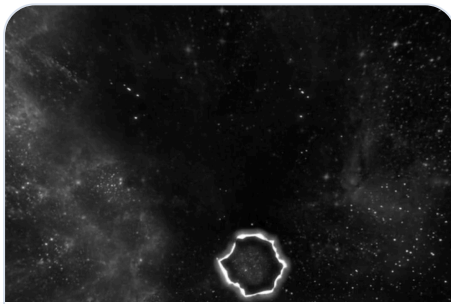
Neostalgia



XR EXHIBITION / VISITOR MEMORY PIPELINE / OFFICIAL SELECTION

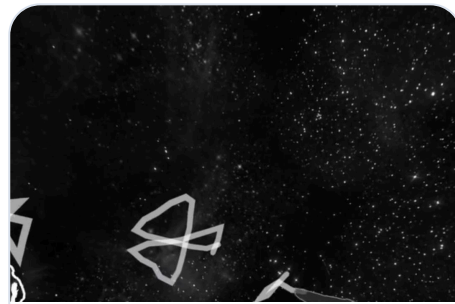
I turned visitor interviews into generated stories and visual assets, then placed them in a Unity XR exhibition. The project was selected by BIFAN XR Beyond Reality and received an MSIT citation.

- I put Whisper interviews, GPT stories, drawing and object assets, and skyboxes into the Unity XR runtime.
- I built the visitor input screens, XR transitions, dashboard, and on-site controls.
- Operators could check generation status and failure reasons from the dashboard during the exhibition.



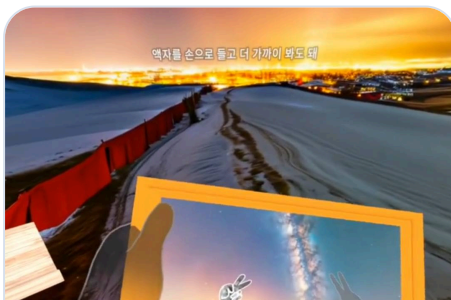
SCREENSHOT

Dialogue and input screen for collecting visitor memories during the exhibition.



SCREENSHOT

Visitor drawing material used as part of the memory-based exhibition pipeline.



SCREENSHOT

3D object asset placed into the XR scene composition.

Oracluna Tarot

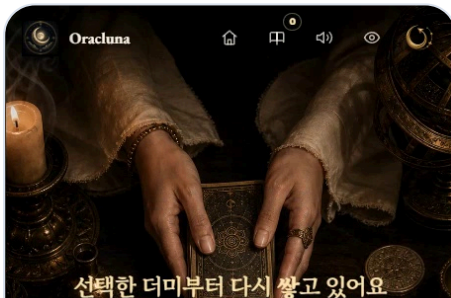


MOBILE-FIRST WEB APP / 78-CARD WHEEL / QDRANT RAG / DOCKER

oraculuna.com

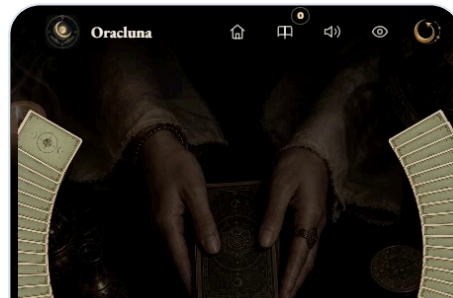
At oraculuna.com, users choose a concern, shuffle and cut the deck, pick from a 78-card wheel, receive an interpretation, and save or share the result. The flow uses one, three, or seven cards according to the concern.

- Built a draggable 78-card wheel with direct card selection and separate slots for chosen cards. Card spacing and scale are calculated so one-, three-, and seven-card spreads stay inside the mobile viewport.
- Built a Qdrant-backed RAG pipeline combining the question, spread position, card, orientation, and retrieved knowledge, with provider adapters for local Ollama development and OpenAI API deployment.
- I automated checks for 78 PNGs, 78 SVGs, 78 metadata files, and 170 retrieval records, and handled iOS Web Audio, pre-generated share posters, and Docker/NAS deployment.



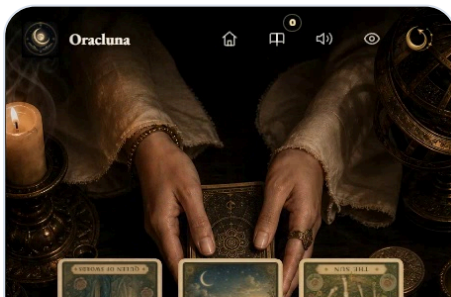
SCREENSHOT

Three cut piles recombining from the pile selected by the user.



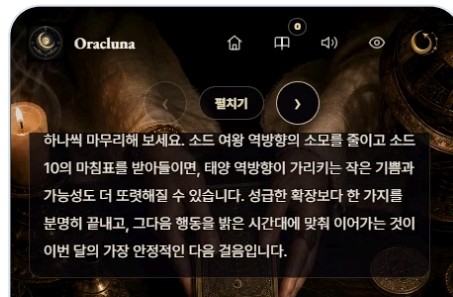
SCREENSHOT

Draggable 78-card wheel with direct selection from the card faces.



SCREENSHOT

Three-card monthly reading showing upright and reversed orientation.



SCREENSHOT

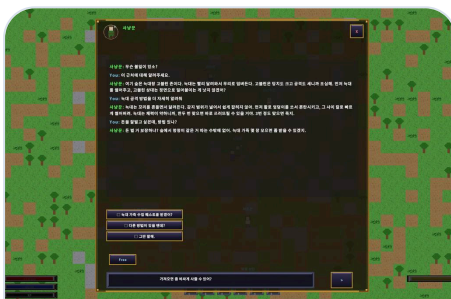
Completed reading with save, share, and restart actions.

GenWorld

UNITY 2D / NPC STATE SYSTEM / OFFLINE PLAY

A Unity 2D RPG prototype with NPC state, dialogue, skills, and offline play. GenWorld falls back to its rule-based dialogue graph when the optional LLM service is unavailable.

- Connected LLM dialogue to a rule-based graph that takes over when the optional service is unavailable.
- Modeled NPC mood, dialogue, skills, and world state as runtime systems inside Unity.
- Used deterministic rules and runtime state inspection to keep the prototype playable offline.



SCREENSHOT

Dialogue flow where NPC response is driven by local context and runtime rules.



SCREENSHOT

NPC mood and state view used to check behavior changes.



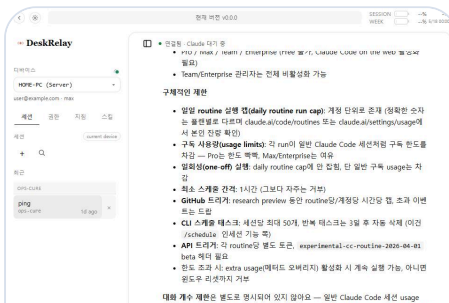
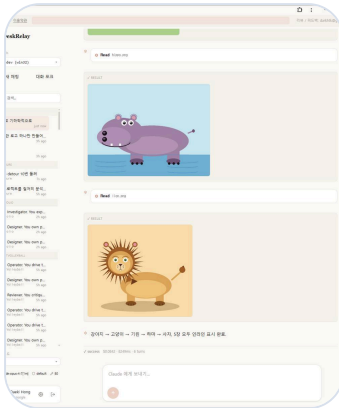
SCREENSHOT

Skill and behavior view for checking rule-based NPC actions.

INTERNAL TOOL / BROWSER AND MOBILE CONTROLS

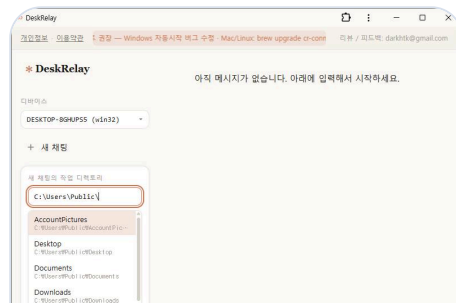
A tool for controlling local work sessions from a browser or phone. It includes device registration, session continuity, diagnostics, permissions, and updates.

- Built the controls for multiple devices, local execution, connection diagnostics, and daily use.
- Built desktop and mobile screens for chat, workspace selection, device switching, settings, permissions, and updates.
- I also built the connection and update settings needed to use it every day.



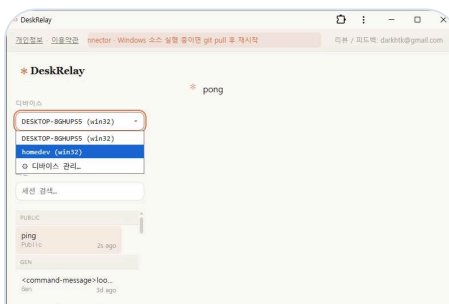
SCREENSHOT

Session screen captured from a working workspace conversation.



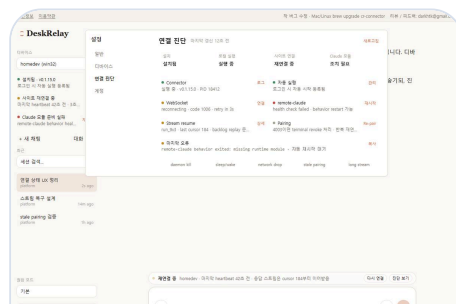
SCREENSHOT

Directory picker flow for choosing a local project safely.



SCREENSHOT

Device switcher showing how one UI can route control to multiple registered machines.

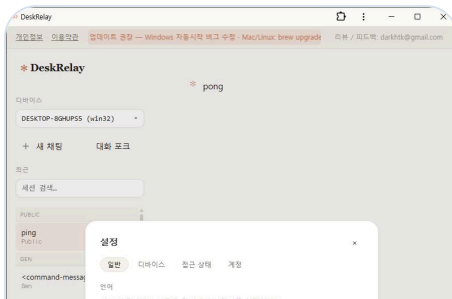


SCREENSHOT

Connection status screen for checking the local server and device channel.

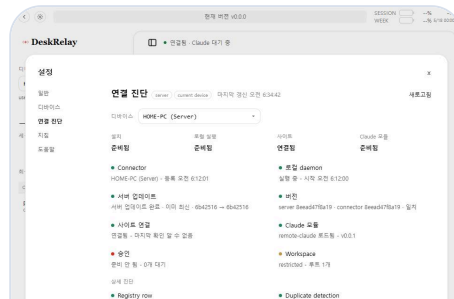
DeskRelay - additional screenshots 1-6 of 8

A tool for controlling local work sessions from a browser or phone. It includes device registration, session continuity, diagnostics, permissions, and updates.



SCREENSHOT

Settings screen for changing runtime behavior.



SCREENSHOT

Diagnostics screen for checking failures without diving into logs first.



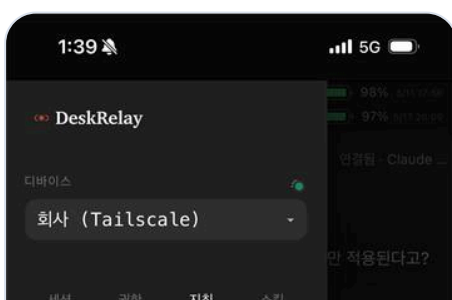
SCREENSHOT

Update settings used for daily maintenance.



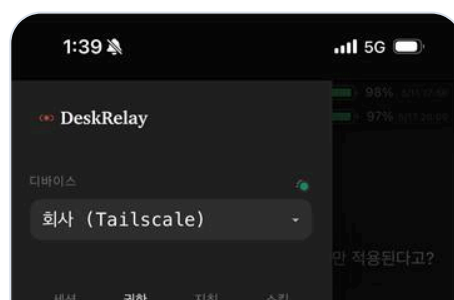
SCREENSHOT

Mobile chat flow for continuing the same coding session away from the desk.



SCREENSHOT

Mobile instructions tab for keeping project guidance visible.

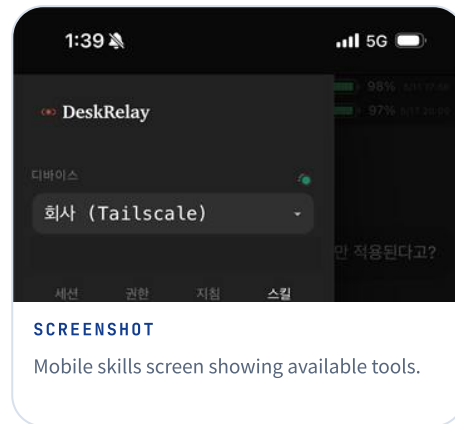
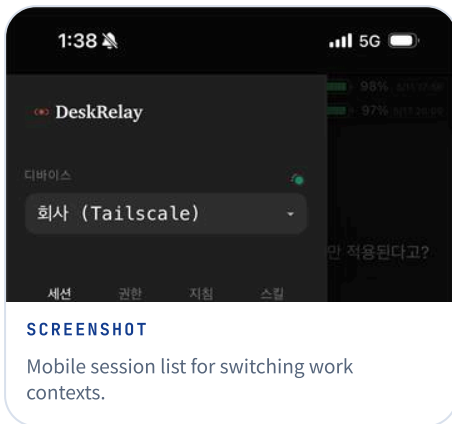


SCREENSHOT

Mobile permissions screen showing allowed capabilities.

DeskRelay - additional screenshots 7-8 of 8

A tool for controlling local work sessions from a browser or phone. It includes device registration, session continuity, diagnostics, permissions, and updates.



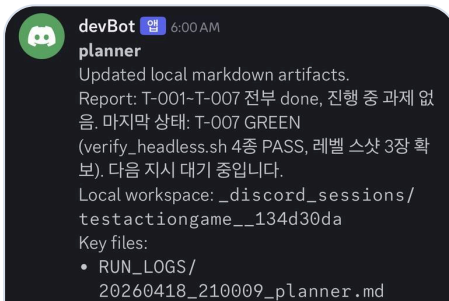
Ops-Cure Workflow



DISCORD OPERATIONS / TASK STATE / WORKFLOW LOOP

Discord threads store the plan, execution notes, current status, and handoff for each task.

- Stored task state and ownership so progress remains visible when work moves between threads.
- Displayed planning, work, checks, and handoff as named states in Discord.
- The Discord thread shows task state, messages, approvals, and checks.



SCREENSHOT

Discord thread showing task state, messages, approvals, and checks.